Kaua‘i – Hurricane ESF-6 playbook

Data as of 2026-10-28

County: Kaua‘i · **Hazard:** Hurricane

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

People at risk **73,511**	Shelter demand (central) **2,205**	Shelter demand low – high **1,470 – 3,676**
Limited English (C) **2,059**	Power-dependent (M) **374**	Disability (I) **8,092**
Age 65+ (I) **15,557**	People without a vehicle (T) **3,184**	Known 2026 sites usable **5**

Shelter-seeking rates: low 2%, central 3%, high 5.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

2. Scope: county total vs. one community

County total = upper bound. It assumes every at-risk tract is hit at once (all tracts). Real events usually hit fewer communities.

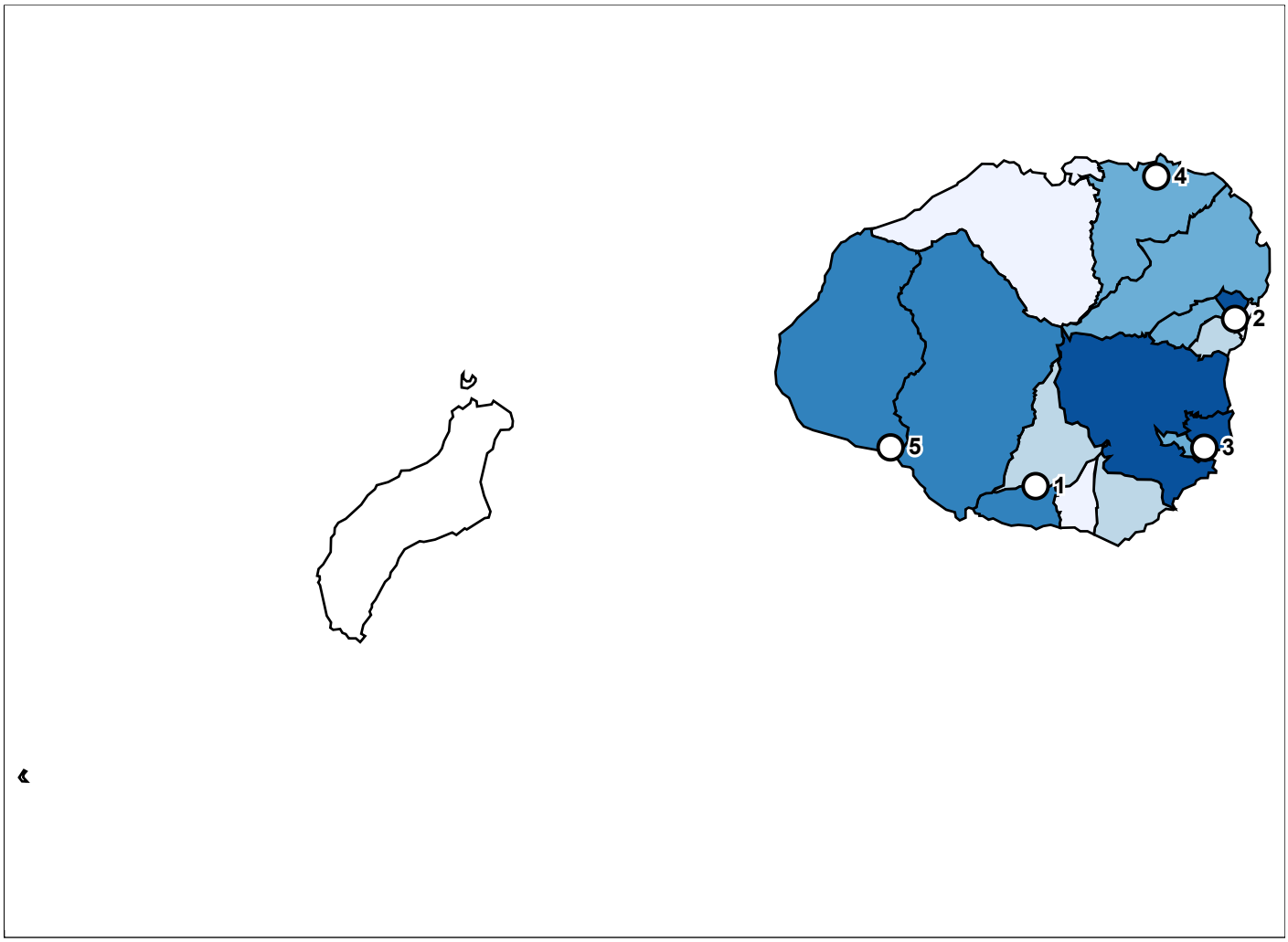
Largest single community (highest-demand census tracts)

Hurricanes affect the whole county, so these are simply the most populous at-risk tracts.

Tract	At risk	Shelter low / central / high	Limited English	Power-dep.	Disability	65+	No vehicle
Tract 404.01 · near Puhi	7,505	150 / **225** / 375	490	25	723	1,048	470
Tract 405 · Lihue	7,301	146 / **219** / 365	276	33	856	1,388	120
Tract 403.01 · Kapaa	6,116	122 / **183** / 306	112	30	872	1,228	565

Plan a realistic one-community event with the first row.

3. Map: AFN need and known sites



- AFN need 0–416 people (class 1 of 5)
- AFN need 416–636 people (class 2 of 5)
- AFN need 636–818 people (class 3 of 5)
- AFN need 818–1,176 people (class 4 of 5)
- AFN need 1,176–1,708 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

4. Known sites (opened in 2026 events)

#	Status	Site	Address	Reason	Times opened 2026
1	✓ **Usable**	Kalāheo Neighborhood Center	4480 Papalina Road, Kalāheo	no site-level screen for this hazard	1
2	✓ **Usable**	Kapa‘a Middle School	4867 Olohena Road, Kapa‘a	no site-level screen for this hazard	1
3	✓ **Usable**	Kaua‘i High School Gym	3577 Lala Road, Līhu‘e	no site-level screen for this hazard	1
4	✓ **Usable**	Kīlauea Neighborhood Center	2460 Keneke Street, Kīlauea	no site-level screen for this hazard	1
5	✓ **Usable**	Waimea Canyon Middle School	9555 Huakai Road, Waimea	no site-level screen for this hazard	1

Candidate public schools in this county usable for this hazard: **20**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

Need	Quantity	Unit	Proposed lead	Proposed support	Pledged	Gap
Cots / sleeping space	2,206	cots	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤72 h)	44,107	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤2 weeks)	88,214	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Water	2,206	gal/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Meals	4,411	meals/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Toilets (≤72 h)	56	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Toilets (≤2 weeks)	111	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Shelter staff	154	staff per shift	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community	—	confirm with partner
Power outlets for power-dependent residents	12	outlets	doh	county_em; health_care; utilities	—	confirm with partner
Accessible transport	3,184	people in no-vehicle households at risk	county_em	dhs; hiema; dot; business_community; other_agencies	—	confirm with partner
Pet space – dog households	308	households with a dog	county_em	hawaii_voad	—	confirm with partner
Pet space – cat households	180	households with a cat	county_em	hawaii_voad	—	confirm with partner
Children's area	474	children in shelter	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy	—	confirm with partner
Interpreters – Ilocano	26	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	14	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Spanish	5	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Japanese	4	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Marshallese	2	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Hawaiian	1	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Central Pacific hurricane season start (June 1) NOAA Central Pacific Hurricane Center	Confirm partner capacity pledges (cots, meals, generators, accessible vehicles, interpreters) and the list of sites usable for wind; refresh priority-outreach tracts.	County Emergency Management (each county)	dhs, hiema, red_cross
BEFORE	Hurricane Watch NOAA NWS / Central Pacific Hurricane Center	Contact priority tracts (power-dependent, no-vehicle, limited English); confirm which shelters will open, generator power for medical devices, and accessible transport.	County Emergency Management (each county)	doh, utilities, dot, dhs
BEFORE	Hurricane Warning NOAA NWS / Central Pacific Hurricane Center	Open shelters including AFN-capable sites; start accessible transport for residents without vehicles; send pre-approved multilingual messages.	County Emergency Management (each county)	red_cross, dhs, doe, dot
AFTER	Warning cancelled or all clear County / NWS / PTWC	Feeding, re-power for medical devices, recovery sheltering for displaced households; coordinate donations and volunteers.	American Red Cross	hawaii_voad, salvation_army, faith_based, utilities

7. Priority outreach and languages

Top 10 census tracts by AFN need (tract-level counts only)

#	Tract	At risk	Power-dep.	Disability	Limited English	No vehicle	AFN need
1	Tract 404.01 · near Puhi	7,505	25	723	490	470	1,708
2	Tract 403.01 · Kapaa	6,116	30	872	112	565	1,579
3	Tract 405 · Lihue	7,301	33	856	276	120	1,285
4	Tract 408 · rural Kaua‘i	4,091	18	544	463	151	1,176
5	Tract 407.02 · near Kalaheo	5,494	28	496	173	383	1,080
6	Tract 409 · rural Kaua‘i	5,172	26	410	140	365	940
7	Tract 9400 · near Kapaa	4,485	22	553	49	193	818
8	Tract 404.02 · Lihue	2,798	21	419	70	249	759
9	Tract 401.05 · near Kilauea	4,508	25	525	94	83	726
10	Tract 402.04 · Wailua Homesteads	6,039	32	558	48	0	638

Languages: people who speak English less than "very well" (Kauai + Maui)

Kaua‘i and Maui share one language list: ACS PUMS reports both counties (and Kalawao) as a single area, PUMA 00100, so they cannot be separated.

Language	People 5+	90% MOE
Ilocano	9,131	±1,439
Tagalog, Filipino	5,074	±926
Spanish	1,711	±554
Japanese	1,339	±417
Marshallese	633	±488
Hawaiian	515	±289

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6

Source	Fetched	URL / file
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
Census TIGERweb 2020 Census Designated Places (tract community labels)	2026-10-27	https://tigerweb.geo.census.gov/arcgis/rest/services/TIGERweb/tigerWMS_Census2020/MapServer/28
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.